

Gaussian SVP: implementation and simulation review

15 September 2026

Scope and mathematical target

For independent observations $Y_i \sim \mathcal{N}(\mu_i, \sigma^2)$, write $y_{s..t} = (y_{s+1}, \dots, y_t)$ and

$$\mathcal{C}(s, t) = \sum_{i=s+1}^t (y_i - \bar{y}_{s..t})^2, \quad f(s, t) = \frac{1}{2\sigma^2} \max_{s < u < t} \{\mathcal{C}(s, t) - \mathcal{C}(s, u) - \mathcal{C}(u, t)\}.$$

`SVP(..., test="gaussian_mean")` uses $\sigma^2 = 1$ and the half log-likelihood-ratio scale above. The argument **gamma** is a **validity threshold**, not the additive penalty in PELT. In the simulations it is $\gamma = c \log n$, where c is the free calibration constant printed in every SVP legend.

For boundaries $0 = \tau_0 < \dots < \tau_K = n$, define the mode-dependent segment statistic

$$F_0(s, t) = f(s, t), \quad F_R(s, t) = \max_{s < u \leq t} f(s, u), \quad F_B(s, t) = \max_{s \leq a < b \leq t} f(a, b).$$

Here F_0, F_R, F_B correspond respectively to **none**, **right**, and **both**. Singleton segments are accepted. With strict validity $F_M(s, t) < \gamma$, each mode solves

$$\hat{\tau}_M \in \arg \text{lex min}_{\substack{0=\tau_0 < \dots < \tau_K=n \\ F_M(\tau_{j-1}, \tau_j) < \gamma, \ j=1, \dots, K}} \left(K, \sum_{j=1}^K \mathcal{C}(\tau_{j-1}, \tau_j) \right).$$

Thus the modes are three related statistical procedures, not three generally interchangeable implementations. For a fixed γ , $F_0 \leq F_R \leq F_B$, so the feasible sets are nested and the optimal segment counts satisfy $K_0 \leq K_R \leq K_B$. Equality of returned endpoints requires the same validity definition and tie rule; calibration to the same null error rate does not make the segmentations identical.

With $R_t = (K_t, Q_t)$ and $R_0 = (0, 0)$, the dynamic program is

$$R_t = \text{lex min}_{\substack{0 \leq s < t \\ F_M(s, t) < \gamma}} \{R_s + (1, \mathcal{C}(s, t))\}.$$

The compiled code compares SSE strictly and scans surviving boundaries in increasing order, so an exact numerical tie keeps the earliest visited boundary.

What the Gaussian implementation does

Each candidate boundary s owns an incremental test state, the last endpoint already supplied to that state, and a validity flag. Arrays **K**, **Q**, and **previous** store R_t and allow backtracking from n . Centered first- and second-moment prefix sums evaluate $\mathcal{C}(s, t)$ in constant time. **GaussianFocusState** maintains two functional-pruning envelopes and returns the maximum Gaussian gain over split points. Its expected number of active pieces is logarithmic under the usual independent continuous model, while its worst case is linear in the candidate-segment length.

The three implementation shortcuts are exact for their stated validity rule:

- **none** keeps all boundaries, groups them by K_s , and tests groups in increasing K_s . It stops after the first group containing a valid candidate, because larger groups cannot improve the primary coordinate.
- **right** returns immediately while boundary zero remains valid. Later candidates are deferred; if zero fails, their missed prefix history is replayed exactly. Afterwards invalid candidates are deleted and SSE is skipped whenever $K_s + 1$ already exceeds the best segment count.
- **both** scans candidates from newest to oldest. At the newest invalid candidate it stops and deletes that candidate and every older one. It first finds the smallest surviving $K_s + 1$, then computes SSE only in that class.

There is no whole-series preflight. In particular, a no-change series does not make **both** return before the dynamic-programming loop.

The other Gaussian-model entries in `SVP()` share the same lexicographic engine but not the same validity statistic:

test	model	one candidate of length m
gaussian_mean	independent mean changes, fixed variance 1	expected $O(\log m)$, worst $O(m)$
gaussian_variance	zero-mean variance changes	$O(a_m)$, worst $O(m)$
AR1	AR(1) mean changes, fixed rho and innovation variance	$O(m)$
AR1Profile	exact AR(1), innovation variance profiled	$O(m)$
AR1Focus	approximate AR(1) innovation transform	$O(a_m)$, worst $O(m)$

AR1Focus can differ from **AR1** even with known parameters because its innovation split omits the exact boundary transition. **AR1Profile** can also differ from fixed-variance **AR1**. All variants currently use ordinary SSE as the secondary tie-breaker, even when that is not their model likelihood.

Correctness audit: changes recommended in the package

The Gaussian mean statistic is mathematically translation invariant, but its FOCuS state currently accumulates the raw observations. Centering is applied only to the SSE prefix sums. The following deterministic check shows the problem (the rows report segment endpoints):

mode	offset	endpoints
none	0e+00	48
none	1e+06	48
none	1e+09	6, 48
none	1e+12	12, 48
right	0e+00	48
right	1e+06	48
right	1e+09	5, 10, 15, 22, 30, 38, 40, 48
right	1e+12	1, 6, 12, 16, 20, 23, 27, 31, 37, 45, 48
both	0e+00	48
both	1e+06	48
both	1e+09	3, 6, 8, 11, 12, 13, 15, 20, 22, 26, 29, 32, 33, 36, 40, 44, 45, 48
both	1e+12	3, 6, 8, 9, 12, 14, 18, 20, 23, 26, 27, 29, 30, 31, 36, 39, 41, 44, 48

The repair should center the observations **before they are passed to both the validity state and the prefix-sum cost calculation**, preferably in the Gaussian-mean dispatch path. Add exact endpoint and predecessor tests for positive and negative offsets $10^2, 10^6, 10^9, 10^{12}$ in every mode. The existing 10^6 check is not sufficiently adversarial.

Three further points require either a code change or an explicit paper qualification:

1. **GaussianVariance** sends Y_i^2 to **GammaRate(shape=1)**. If $Y_i \sim \mathcal{N}(0, \sigma^2)$, then Y_i^2 is Gamma with shape $1/2$. The constructor should use shape $1/2$, followed by null and power tests.
2. Validity is implemented as **statistic < gamma**; equality fails. The paper must use the same strict inequality, or the implementation and boundary tests must be changed together.
3. Proposition 2 applies to the final-segment rule **none** on the stated cost scale. It does not automatically justify a **right** or **both** curve, and a PELT SSE penalty and the half-likelihood SVP threshold must not be called the same parameter merely because both contain $2 \log n$.

The public default is presently **both**. Therefore, the paper’s basic SVP definition agrees with the software default only if it uses F_B ; otherwise the paper should request **subtests="none"** explicitly. No package behavior was changed in this review.

Recalibration and Section 5.1

The calibration target is the probability of at least one internal false change under $n = 1000$ independent standard Gaussian observations. PELT uses its Manual penalty $2 \log n$. For each SVP mode, a grid search chooses c in $\gamma = c \log n$ to match PELT on 20,000 calibration series; 10,000 new series are used only for validation.

mode	c	calibration FPR	validation FPR	PELT validation FPR	absolute validation gap
none	0.968	0.0158	0.0141	0.0144	0.0003
right	1.642	0.0159	0.0161	0.0144	0.0017
both	1.987	0.0159	0.0155	0.0144	0.0011

The validation Wilson intervals overlap for all three modes. Section 5.1 keeps four curves: PELT, SVP Gaussian / right / $c = 2.000$, calibrated right, and calibrated both. Calibrated none is used in the runtime study only. Every series is generated once and shared across methods.

The scoring code now removes the mandatory endpoint n before one-to-one tolerance matching. The earlier code treated n as a true and detected change, inflating F1. For example, the mean PELT F1 over the jump grid changes from 0.658 to 0.591 in **rand1**, although the fitted endpoints are identical. PELT and right, $c=2$ reproduce exactly after rescoring. Recalibration changes the two calibrated fits; its largest F1 change after holding the scoring rule fixed is small in a change scenario.

stage	configuration	none	rand1	up	updown
historical fits, corrected score	PELT	0.987	0.591	0.565	0.545
historical fits, corrected score	both calibrated	0.953	0.577	0.542	0.520
historical fits, corrected score	right $c=2$	1.000	0.555	0.526	0.473
historical fits, corrected score	right calibrated	0.964	0.563	0.543	0.508
refreshed calibration and corrected score	PELT	0.987	0.591	0.565	0.545
refreshed calibration and corrected score	both calibrated	0.983	0.575	0.538	0.505
refreshed calibration and corrected score	right $c=2$	1.000	0.555	0.526	0.473
refreshed calibration and corrected score	right calibrated	0.983	0.562	0.536	0.498

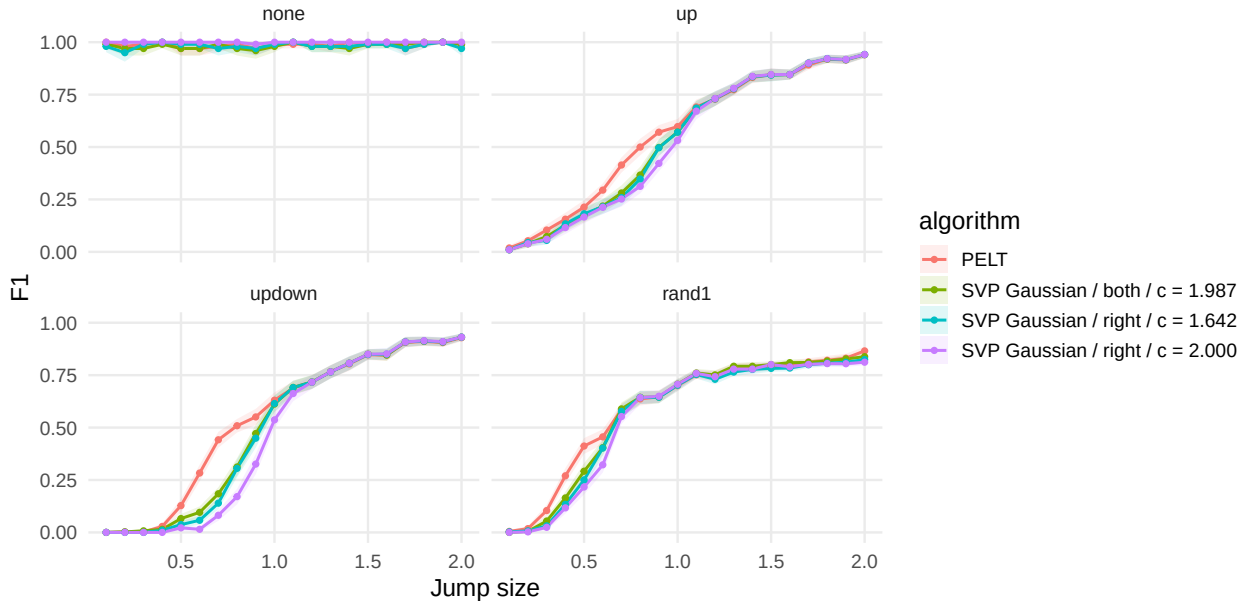


Figure 1: Refreshed Section 5.1 F1 curves. SVP labels report the test name, subtests mode, and free constant.

The revised conclusion is more cautious than the current paper. PELT has the best average F1 in the three change scenarios. The calibrated SVP curves are close to PELT in **rand1** around jump 0.9 and in **up/updown** near jump 2, but they remain conservative at small and moderate jumps. For **rand1** at jump 2, mean F1 is about 0.87 for PELT and 0.84 for calibrated both, so the methods are not uniformly indistinguishable above jump 0.9.

Figure 5: time and complexity

The refreshed benchmark uses 20 logarithmically spaced sizes from 256 to 16,384 with 20 replications under no change, and $n = 10,000$ with 0–100 true changes and 10 replications. Data come from `ts_generator()`. Methods receive the same series in randomized order, after warm-up. Calls shorter than 0.05 s are batched and divided by the batch size. Curves show median elapsed time; bands are nonparametric bootstrap intervals.

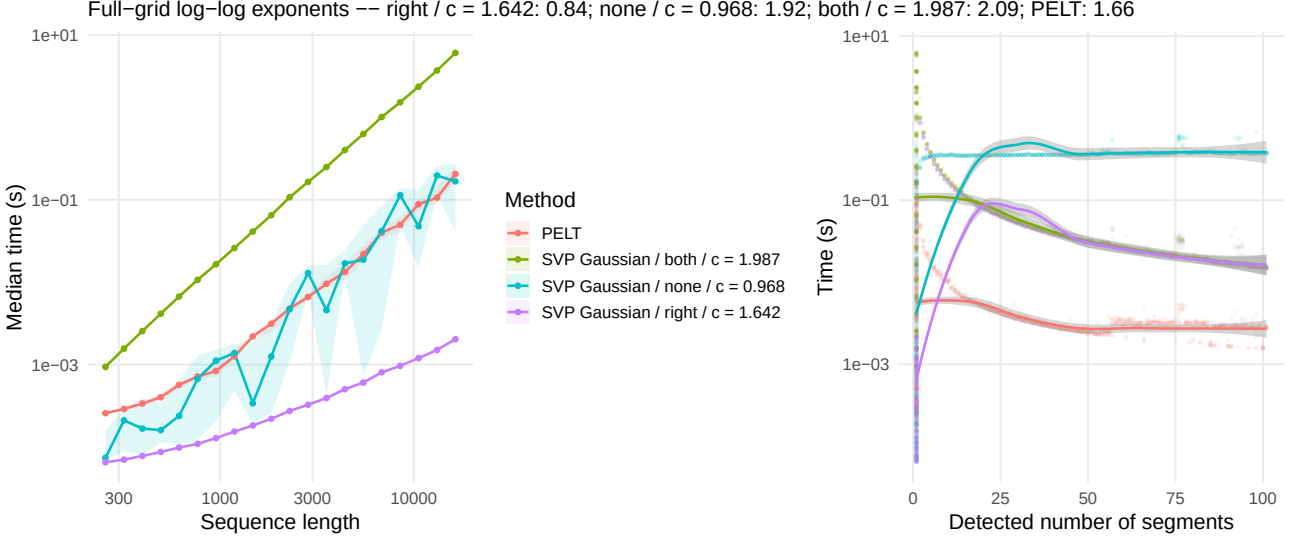


Figure 2: Controlled Gaussian runtime comparison. The exponents printed in panel (a) are descriptive log-log fits, not worst-case bounds.

method	range	exponent	2.5%	97.5%
SVP Gaussian / right / c = 1.642	full grid	0.840	0.834	0.843
SVP Gaussian / right / c = 1.642	upper half	1.013	0.992	1.023
SVP Gaussian / none / c = 0.968	full grid	1.915	1.683	2.035
SVP Gaussian / none / c = 0.968	upper half	1.923	1.414	2.460
SVP Gaussian / both / c = 1.987	full grid	2.090	2.086	2.094
SVP Gaussian / both / c = 1.987	upper half	2.044	2.032	2.051
PELT	full grid	1.665	1.640	1.702
PELT	upper half	1.910	1.835	2.014

Let E_t be the number of candidate validity states actually checked at endpoint t . Gaussian running time is described by $\sum_t O(E_t \log t)$ in the usual expected FOCuS regime, with a larger worst case when active functional pieces grow linearly. The shortcuts determine E_t :

- Under no change, **right** usually tests only boundary zero, hence expected $O(n \log n)$ validity work. The historical sub-quadratic SVP curve was also a prefix-validity implementation with this one-segment behavior; its slope near 1.4 was empirical, affected by fixed overhead, coarse millisecond timings, and a short finite range. It was not evidence for a universal $n^{1.4}$ complexity.
- **none** stops in the $K_s = 0$ bucket only while boundary zero is valid. Once that candidate fails, the $K_s = 1$ bucket can contain many boundaries and all valid candidates in that bucket must be compared for the SSE tie-breaker. At the calibrated threshold these episodes produce a noisy, nearly quadratic curve (full-grid exponent about 1.92); retaining candidates lets boundary zero recover, but does not erase the cost already incurred.
- **both** has no zero-boundary shortcut. If all recent candidates are valid, the reverse scan examines them all. Thus the no-change candidate exposure is quadratic and the FOCuS factor gives expected $O(n^2 \log n)$ in this regime. It becomes advantageous only when a recent failure deletes a long prefix of candidates.

The old whole-series preflight had hidden this cost by returning the one-segment answer before dynamic programming; it has been removed. Figure 5 must therefore not claim that **both** is uniformly the fastest or that its pruning caps the candidate count under no change. PELT’s observed slope is likewise implementation- and regime-specific; its standard pruning gives linear expected time under assumptions, although adverse cases can be quadratic.

Required edits to SVP_NEW.pdf

Section 5.1 should (i) call γ a validity threshold, (ii) define $\gamma = c \log n$, (iii) state the four plotted methods and calibrated constants, (iv) describe the null-FPR calibration/validation samples, and (v) define F1 using internal changepoints and one-to-one matching. Replace the claim of near-equivalence above jump 0.9 by the result-specific comparison given above.

Section 5.4 should replace its three historical SVP configurations by calibrated **none**, **right**, and **both**; state the controlled timing protocol; and distinguish empirical exponents from bounds. The explanation of **both** must reflect the absent preflight and its quadratic no-change candidate scan. The Figure 5 caption should identify medians, bootstrap intervals, hardware, and the exact package versions.

Finally, revise the general method section so the base feasibility rule and the software **subtests** value agree; qualify Proposition 2 as applying to the **none** rule and the correct objective scale; use strict validity consistently; and correct the statement that the **changepoint** PELT implementation is based on Rcpp. A standalone LaTeX replacement memo accompanies this document.